

Dark Pattern Guardian

A 100% Built From Scratch Solution for
Fintech & Consumer Protection.

 Code2Create

 Team Infinite Loop



The Guardian Extension

Dark Pattern Guardian is a purpose-built Chrome Extension designed as a digital shield against e-commerce manipulation.

- 🛡️ **Seamless Integration:** Resides quietly in your browser toolbar without disrupting standard navigation.
- 🛡️ **Autonomous Scanning:** Silently analyzes page text during checkout flows to identify manipulative tactics.
- 🛡️ **Instant Alerts:** Provides color-coded UI warnings when fake urgency or hidden fees are detected.
- 🛡️ **100% Private:** Runs locally on your device—your browsing habits and data never leave your machine.



Problem Statement & Context



The Crisis

Over 76% of websites and apps examined in a 2024 ICPEN review used at least one possible dark pattern, including urgency and hidden-information tactics.



Consumer Impact

Consumers frequently face rushed decisions, unintended purchases, privacy compromises, and unexpected costs due to manipulative UI design.



Current Gaps

Existing ad blockers and anti-trackers miss semantic UI manipulation, ultimately failing to catch predatory copy, confirmshaming, and subtle drip pricing.

Proposed Solution & Novelty



Zero External APIs

Entirely free from commercial wrapper libraries or third-party web scrapers. We built a fully independent processing pipeline to ensure total control.



Lightweight MV3 Bridge

A custom Chrome extension leveraging strictly activeTab permissions to scan page text securely without persistent background bloat.



Proprietary Heuristic Parser

Powered by a custom-coded Python regex engine and a dynamic threat-scoring matrix designed from the ground up for absolute precision.

Architecture & Workflow

1 DOM Extraction

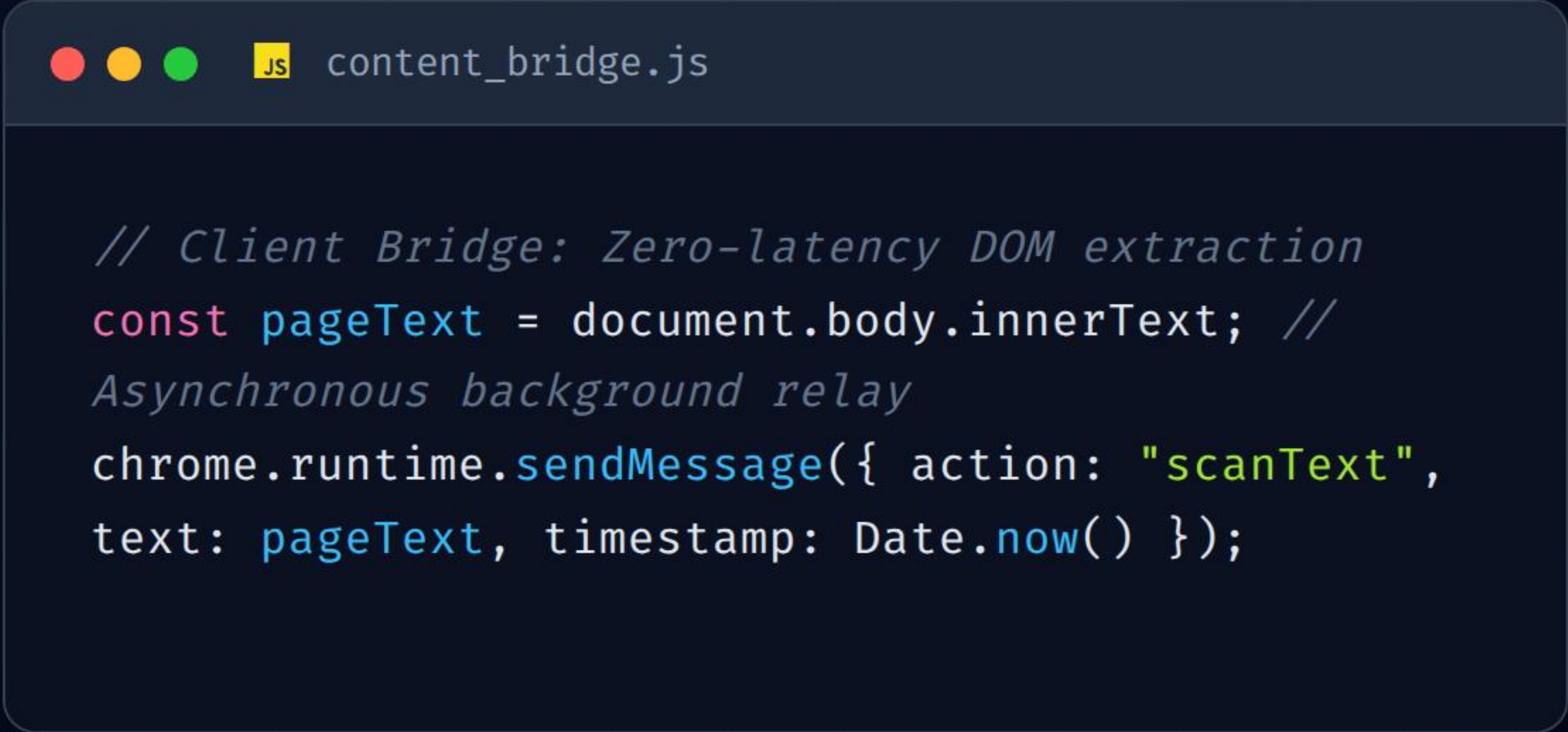
Injects lightweight client script upon page load to extract text payload instantly from active page body.

2 Secure Transmission

Background event handler relays extracted payload via zero-latency messaging bridge to Local Flask backend server.

3 Heuristic Analysis

Python backend evaluates text stream against custom regex pattern dictionaries to calculate risk metrics and threat scores.



```
// Client Bridge: Zero-latency DOM extraction  
const pageText = document.body.innerText; //  
Asynchronous background relay  
chrome.runtime.sendMessage({ action: "scanText",  
text: pageText, timestamp: Date.now() });
```


Feasibility & Implementation



High Performance

Extracting relevant page text instead of raw HTML significantly reduces unnecessary data transfer and processing overhead, maintaining native browsing speeds.



Privacy-Focused Design

The MVP is designed for stateless analysis: page text is processed strictly for the immediate scan without storing it as persistent user data on servers.



Scalability

The modular Python architecture allows seamless future scaling into advanced machine learning models and open-source zero-shot transformers.

Competitive Advantage

Capability	Standard AdBlockers	Anti-Trackers	Dark Pattern Guardian
Detects Fake Scarcity & Urgency Timers	✗ No	✗ No	✓ Yes (Custom Engine)
Unmasks Hidden Drip Fees & Confirmshaming	✗ No	✗ No	✓ Yes (Heuristic Weights)
Built 100% From Scratch (Zero Black-Box APIs)	✗ No	✗ No	✓ Yes (Full Compliance)

Team Infinite Loop



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Image Sources



https://img.magnific.com/free-photo/digital-cybersecurity-shield-network-background_84443-85964.jpg?semt=ais_hybrid&w=740&q=80

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Source: elements.envato.com